# Week 6 - Graded Assignment 6

**The due date for submitting this assignment has passed.**
Due on 2026-03-25, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.

## You have last submitted on: 2026-03-25, 07:24 IST

**Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.**

**Multiple Select Questions (MSQ):**

*1 point*
A function $T: V \rightarrow W$ between two vector spaces $V$ and $W$ is said to be a linear transformation if the following conditions hold:

- Condition 1: $T(v_1 + v_2) = T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$.
- Condition 2: $T(cv) = cT(v)$ for all $v \in V$ and $c \in R$.

Consider the following function:

$$T: R^2 \rightarrow R$$
$$T(x, y) = \begin{cases} 3x & \text{if } y = 0 \\ 4y & \text{if } y \neq 0 \end{cases}$$

Which of the following statements is true?
- [ ] Condition 1 holds.
- [x] Condition 1 does not hold.
- [x] Condition 2 holds.
- [ ] Condition 2 does not hold.

## Yes, the answer is correct.
## Score: 1
## Accepted Answers:
Condition 1 does not hold.
Condition 2 holds.

*1 point*
Suppose the matrix representation of a linear transformation $T: R^3 \rightarrow R^3$ with respect to ordered bases $\beta = \{(1, 0, 1), (0, 1, 0), (0, 0, 1)\}$ for the domain and $\gamma = \{(1, 0, 0), (0, 1, 0), (1, 0, 1)\}$ for the range, is $I_{3\times 3}$, i.e., the identity matrix of order 3. Let $A$ denote the matrix representation of the linear transformation $T$ with respect to the standard ordered basis of $R^3$ for both domain and range. Which of the following are true?

- [ ] $A = I_{3\times 3}$ i.e., identity matrix of order 3.
- [ ] $A$ is a singular matrix
- [ ]

$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$A = $\begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

- [x] $det(A)$ = 1det(A) = 1.
- [ ] $det(A)$ = $-$1det(A) = −1.
- [x] $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$A = $\begin{bmatrix} 0 & 0 & -1 & 0 & 1 & 0 & 1 \\ 0 & 0 & -1 \end{bmatrix}$

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
$det(A)$ = 1det(A) = 1.
$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$A = $\begin{bmatrix} 0 & 0 & -1 & 0 & 1 & 0 & 1 \\ 0 & 0 & -1 \end{bmatrix}$

*1 point*
Match the linear transformations and sets of vectors in column A with the images of those sets under the linear transformation in column B and the geometric representations of both sets in column C.

Choose the correct option from the following.

- [ ] i $\rightarrow$ $\rightarrow$ d $\rightarrow$ $\rightarrow$ 2, ii $\rightarrow$ $\rightarrow$ b $\rightarrow$ $\rightarrow$ 4.
- [x] i $\rightarrow$ $\rightarrow$ b $\rightarrow$ $\rightarrow$ 4, ii $\rightarrow$ $\rightarrow$ d $\rightarrow$ $\rightarrow$ 2.
- [x] iii $\rightarrow$ $\rightarrow$ a $\rightarrow$ $\rightarrow$ 1, iv $\rightarrow$ $\rightarrow$ c $\rightarrow$ $\rightarrow$ 3
- [ ] iii $\rightarrow$ $\rightarrow$ a $\rightarrow$ $\rightarrow$ 3, iv $\rightarrow$ $\rightarrow$ c $\rightarrow$ $\rightarrow$ 1

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
i $\rightarrow$ $\rightarrow$ b $\rightarrow$ $\rightarrow$ 4, ii $\rightarrow$ $\rightarrow$ d $\rightarrow$ $\rightarrow$ 2.
iii $\rightarrow$ $\rightarrow$ a $\rightarrow$ $\rightarrow$ 3, iv $\rightarrow$ $\rightarrow$ c $\rightarrow$ $\rightarrow$ 1
*1 point*
Consider a linear transformation $S: M_2(R) \rightarrow M_2(R)$S : M2(R) $\rightarrow$ M2(R) such that $S(A) = A^T$ S(A) = AT . Let $B$B be the matrix representation of $S$S with respect to the ordered bases:

$$\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\}\{[1000], [0010], [0100], [0001]\}$$

of $M_2(R)$M2(R). Choose the set of correct options:

- [ ] The order of the matrix $B$B is $2 \times 2$2 × 2.

- [x] The order of the matrix $B$B is $4 \times 44 \times 4$.
- [x] The dimension of the row space of the the matrix $B$B is 4.
- [ ] The dimension of the column space of the matrix $B$B is 3.
- [ ] The nullity of the the matrix $B$B is 1.
- [x] The rank of the the matrix $B$B is 4.
- [x] $S$S is surjective.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

The order of the matrix $B$B is $4 \times 44 \times 4$.
The dimension of the row space of the the matrix $B$B is 4.
The rank of the the matrix $B$B is 4.
$S$S is surjective.

*1 point*

Let $T: R^n \rightarrow R^m$T : Rn → Rm be a linear transformation and tet $\{v_1, v_2, \ldots, v_n\}$$\{v1, v2, \ldots, vn\}$ be a basis of $R^n$Rn. Suppose $\{T(v_1), T(v_2) \ldots, T(v_n)\}$$\{T(v1), T(v2) \ldots, T(vn)\}$ is a spanning set for $R^m$Rm. Choose all the correct statements.

- [x] $T$T is a surjective linear transformation.
- [ ] $T$T is a linear isomorphism.
- [x] $m \leq n$m ≤ n.
- [x] For any basis $\{w_1, w_2, \ldots, w_n\}$$\{w1, w2, \ldots, wn\}$ of $R^n$Rn, the set $\{T(w_1), T(w_2), \ldots, T(w_n)\}$ $\{T(w1), T(w2), \ldots, T(wn)\}$ is a spanning set for $R^m$Rm.
- [ ] The matrix of the linear transformation $T$T has rank equal to $n$n.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

$T$T is a surjective linear transformation.
$m \leq n$m ≤ n.
For any basis $\{w_1, w_2, \ldots, w_n\}$$\{w1, w2, \ldots, wn\}$ of $R^n$Rn, the set $\{T(w_1), T(w_2), \ldots, T(w_n)\}$ $\{T(w1), T(w2), \ldots, T(wn)\}$ is a spanning set for $R^m$Rm.

**Comprehension Type Question:**

Suppose a bread-making machine B makes 6 breads from 2 eggs, 3 (in hundreds) grams of wheat, and 1 (in hundred) grams of sugar. B also makes 8 breads from 3 eggs, 4 (in hundreds) grams of wheat, and 2 (in hundreds) grams of sugar, and 10 breads from 5 eggs, 5 (in hundreds) grams of wheat, and 3 (in hundreds) grams of sugar. Suppose the production of breads is a linear function of the amount of eggs, wheat (in hundreds), and sugar (in hundreds) used as raw ingredients. Based on the above data answer the following questions. Suppose $x$x eggs, $y$y (in hundreds) grams of wheat, and $z$z (in hundreds)

grams of sugar are used as the raw materials to produce $ax + by + cz$ ax + by + cz number of breads. We can express this as follows:

$$T: R^3 \rightarrow R \quad \text{T : R3 → R}$$

$$T(x, y, z) = ax + by + cz \quad \text{T(x, y, z) = ax + by + cz}$$

where the co-ordinates in $R^3$ R3 denote the number of eggs, amount (in grams) of wheat (in hundreds), and amount (in grams) of sugar (in hundreds). Observe that $T$ T is a linear transformation.

**Use the above information for Question 6, 7 and 8.**

*1 point*

Choose the correct set of options from the the following. (MSQ)

- [ ] $Nullity(T)$ = 1Nullity(T) = 1
- [x] $Rank(T)$ = 1Rank(T) = 1
- [x] $Nullity(T)$ = 2Nullity(T) = 2
- [ ] $Rank(T)$ = 2Rank(T) = 2
- [ ] $Nullity(T)$ = 3Nullity(T) = 3
- [ ] $Rank(T)$ = 3Rank(T) = 3
- [ ] $T$ T is neither one to one nor onto.
- [ ] $T$ T is one to one but not onto.
- [x] $T$ T is onto but not one to one.
- [ ] $T$ T is an isomorphism.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

$Rank(T)$ = 1Rank(T) = 1

$Nullity(T)$ = 2Nullity(T) = 2

$T$ T is onto but not one to one.

*1 point*

Choose the set of correct statements. (MSQ)

- [ ] If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then 9 breads are produced.
- [x] If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.
- [x] If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.
- [ ] If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then no bread is produced.
- [x] If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

If 4 eggs and 2 (in hundreds) grams of sugar is used, and no wheat is used, then no bread is produced.

If only 3 (in hundreds) grams of wheat is used, then 6 breads are produced.
If 3 (in hundreds) grams of wheat and 1 (in hundred) grams of sugar is used, and no egg is used, then 6 breads are produced.
How many breads are produced by the machine from 6 eggs, 10 (in hundreds) grams of wheat, and 5 (in hundreds) grams of sugar? (NAT)

20

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
(Type: Numeric) 20
*1 point*

Let $T$T be a linear transformation from $R^{10}$R10 to $R^{12}$R12. Suppose a basis for the null space of $T$T has 33 vectors. How many linearly independent vectors are needed to form a basis for the range of $T$T?

7

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
(Type: Numeric) 7
*1 point*

Let $M$M be the set of all skew-symmetric matrices of order 7. Then $M$M forms a vector space under matrix addition and scalar multiplication. Let $T$T be a linear transformation from $M$M to $R$R defined by $T(A)$ = 6 $tr(A)$T(A) = 6 tr(A), where $tr(A)$ = tr(A) =trace of $A$A. What is the nullity of $T$T?

21

**Yes, the answer is correct.**
**Score: 1**
**Accepted Answers:**
(Type: Numeric) 21.0
*1 point*

*1 point*
Let $T: R^3 \longrightarrow R^2$T : R3 ⟶ R2 be defined by $T(x, y, z)$ = $(10x + 9y, 2z)$T(x, y, z) = (10x + 9y, 2z). Choose the correct options about $T$T.

- [ ] The matrix of $T$T is $\begin{bmatrix} 100 \\ 9\ 0 \\ 0\ 2 \end{bmatrix}$ $\begin{bmatrix} \\ \\ 1090002 \end{bmatrix}$.
- [x] The matrix of $T$T is $\begin{bmatrix} 1090 \\ 0\ 02 \end{bmatrix}$[1009002].
- [x] A basis for the null space of $T$T is $\{(1, -\frac{10}{9}, 0)\}$$\{(1, -910, 0)\}$.
- [ ] A basis for the null space of $T$T is $\{(1, -\frac{10}{9}, 0), (0, 0, 2)\}$$\{(1, -910, 0), (0, 0, 2)\}$.
- [ ] A basis for the range of $T$T is $\{(10, 10)\}$$\{(10, 10)\}$.
- [x]

A basis for the range of $T$T is $\{(10, 9), (0, 2)\}$$\{(10, 9), (0, 2)\}$.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

The matrix of $T$T is $[\begin{matrix}10 & 9 \\ 0 & 2\end{matrix}]$[1009002].

A basis for the null space of $T$T is $\{(1, -\frac{10}{9}, 0)\}$$\{(1, -910, 0)\}$.

A basis for the range of $T$T is $\{(10, 9), (0, 2)\}$$\{(10, 9), (0, 2)\}$.